Classical Fourth-Order Runge–Kutta

Four-stage explicit integration

1 Butcher tableau and stages

The classical method has tableau

$$\begin{array}{c|cccc} 0 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ \hline & \frac{1}{6} & \frac{1}{3} & \frac{1}{3} & \frac{1}{6} \end{array}$$

and, for $\dot{z} = f(t, z)$,

$$k_1 = f(t_n, z_n), \tag{1}$$
$$k_2 = f(t_n + h/2, z_n + hk_1/2), \tag{2}$$
$$k_3 = f(t_n + h/2, z_n + hk_2/2), \tag{3}$$
$$k_4 = f(t_n + h, z_n + hk_3), \tag{4}$$
$$z_{n+1} = z_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4). \tag{5}$$

The non-autonomous stage times are part of the model.

2 Order and stability

The tableau satisfies the Runge–Kutta order conditions through order four; the local error is $\mathcal{O}(h^5)$ and the global error is $\mathcal{O}(h^4)$ for sufficiently smooth problems. On $y' = \lambda y$, its stability polynomial is

$$R(\xi) = 1 + \xi + \frac{\xi^2}{2} + \frac{\xi^3}{6} + \frac{\xi^4}{24}.$$

The finite stability region does not make RK4 an A-stable method.

3 Geometric properties

Classical RK4 is not symmetric and is not symplectic for a general Hamiltonian system. Its high local accuracy can nevertheless make short-time energy and area errors small. The distinction is structural: symplecticity is the exact identity

$$(D\Phi_h)^T \Omega D\Phi_h = \Omega,$$

not merely a small defect at one step size. The implementation exposes each complete step map to observers so local and accumulated defects can be measured.

4 Energy-tracking configuration

The public model has an optional `track_energy` configuration. When enabled for dynamics satisfying the extended Hamiltonian contract, RK4 advances the triangular state (z, k) with

$$\dot{k} = -\partial_t H(t, z), \qquad K(z, t, k) = H(t, z) + k.$$

The same RK4 nodes and weights are used for z and k. The physical trajectory is returned normally; k and the maximum sampled drift of K are diagnostics. This extension monitors generalized energy but does not turn RK4 into a symplectic or energy-preserving method.

5 Implementation correspondence

The method is implemented in `src/simulation/methods/classical/rk4.py` and consumes the generic dynamics protocol. Main-grid steps are independent of the output schedule; shadow steps provide off-grid samples without changing the accepted trajectory or emitting observations. See the sibling runtime document for the exact result and diagnostic contract.